

Scaling ML-KEM to (much) higher security

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1 Introduction

After a multi-year, multi-round public competition, NIST published its first post-quantum standards FIPS-203 [16], FIPS-204 [15], and FIPS-205 [17] in 2024, specifying ML-KEM, ML-DSA, and SLH-DSA respectively. Interestingly, four of the schemes in the PQC Round 3 Submissions [14] — Kyber [5] (standardized as ML-KEM), Saber [8], NTRU LPrime [3], and FrodoKEM [4] — are closely related lattice KEMs, spanning a spectrum of algebraic structure, from FrodoKEM’s unstructured LWE, through Kyber’s and Saber’s cyclotomic modules, to the prime-degree number fields of NTRU Prime. This spectrum reflects different design principles, commonly understood as structure vs conservatism. ML-KEM’s module structure gives small keys and fast NTT arithmetic, but is also seen as a potential attack surface, so conservative designs discard it entirely [4]. FrodoKEM, for example, is based on a more conservative hardness assumption of plain LWE, which is why bodies such as BSI, ENISA, and NLNCSA point to FrodoKEM for long-term confidentiality [18]; it was also recently standardized by ISO [11] for this very reason.

It is useful to separate the two notions of conservatism this framing bundles together as they concern two independent (and today, entirely hypothetical) classes of attack. The first arises from the choice of hardness assumption. FrodoKEM rests on unstructured LWE to protect against a structure break, an attack that would exploit the algebraic structure of ring/module LWE, but not apply to plain LWE. The second involves the security margin a scheme carries above its target level, which protects against advances in the best known attacks today. This could include improved lattice reduction, dual attacks, or sharper cost models that would affect the concrete security of every lattice scheme at its current parameters, including FrodoKEM. Both of these hedges are distinct and not to be substituted; structured conservatism does nothing against the second class of attacks and no security margin protects against the first. On top of that, it’s impossible to predict which of the two threats is more likely.

Faced with this uncertainty, increasing the margin has been a historically standard response. For computationally secure schemes, concrete security estimate is the cost of the best attack known at the time, and such estimates evolve

as cryptanalysis matures. For instance, the recommended size of an RSA modulus climbed from 512 to 1024 to 2048 bits and beyond as factoring algorithms and available computation advanced [2]. Lattice estimates are not static either. The cost models used for estimation and the attacks have evolved over more than a decade [7,1], and the cost of the best known attacks continues to drift in both directions as improvements are proposed [10,12], scrutinized [9], and refined [6].

If the best known attacks continue to improve, how much margin is required remains open, and the useful question is not whether today’s KEMs carry enough but how much margin they can carry at all. This question arises because additional margin comes at a cost, requiring larger parameters and hence higher bandwidth and runtime, so no scheme can indefinitely withstand advances in cryptanalysis simply by scaling its parameters. For LWE-family KEMs there is a further constraint, since the parameters that determine security also control the decryption failure probability, so increasing the noise to gain a larger security margin necessarily affects correctness.

Crucially, what structure determines is the cost of buying additional margin. At matched security levels, an unstructured scheme already requires roughly an order of magnitude more bandwidth than a module scheme [4,5]. Since these communication costs are considered practical enough for conservative deployments, a module scheme can spend the same bandwidth budget on larger parameters and obtain a conservative security margin while retaining the efficiency advantages of NTT arithmetic.

Recent work has explored this high-margin regime from the perspective of unstructured LWE. KyFrog [13] increases the LWE dimension and adjusts the noise distribution to achieve substantially higher security estimates for much larger ciphertexts. This motivates the complementary question of how much security margin a module KEM can achieve within the communication budget of a conservative unstructured scheme. We answer this question with KAIBURR, a module KEM that scales ML-KEM’s module rank and introduces a generalized noise distribution to preserve correctness. We show that KAIBURR achieves large security margins within FrodoKEM’s communication cost while running substantially faster.

1.1 Our contributions

1.2 Structure of the paper

1.3 Access to artifacts

All software, including parameter analysis scripts, implementation source code, and other artifacts, is publicly available on GitHub.

2 Preliminaries

3 Warmup: Just changing k

One of the main mechanisms to increase bit security in Kyber is to raise the module rank k . Fixing the ML-KEM base ring, noise distribution, and modulus, and varying only k , we measure the decryption-failure probability using the scripts provided by KYBER team⁵. Throughout we use the centered binomial noise CBD_2 (the noise of ML-KEM-768 and -1024; only ML-KEM-512 uses CBD_3). The ranks $k = 3$ and 4 recover the standardized sets ML-KEM-768, and -1024, while $k \geq 5$ is an extrapolation. Table 1 reports the results. Against the 2^{-128} failure bound expected at this security level, the bound holds at $k = 7$ ($2^{-133.7}$) but fails at $k = 8$ ($2^{-116.8}$), so raising k further cannot buy higher security without sacrificing correctness.

These are computed without ciphertext compression, and the rounding that Kyber applies to $\mathbf{u}$ and v produces additional error that can only increase the failure probability. The true failure probability at each rank is therefore at least as large as shown, and the correctness ceiling on k is if anything tighter than the table suggests, in particular, the margin at $k = 7$ may itself erode once compression is included.

Table 1. Decryption-failure probability and public-key and ciphertext sizes as a function of module rank k , at fixed ML-KEM noise and modulus, without ciphertext compression. Ranks $k = 3, 4$ correspond to ML-KEM-768, and -1024.

k	Failure probability
3	$2^{-299.6}$
4	$2^{-231.1}$
5	$2^{-186.5}$
6	$2^{-155.9}$
7	$2^{-133.7}$
8	$2^{-116.8}$

4 Revisiting the noise distribution

5 Benchmarks

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⁵ <https://github.com/pq-crystals/security-estimates>

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